Road Damage Detection System

Object Detection using YOLOv8

Computer Vision Project Report | 2026

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1. Introduction

Road infrastructure plays a critical role in transportation safety and economic development. Timely detection and classification of road damage — such as cracks and potholes — is essential for maintenance planning. This project applies state-of-the-art deep learning object detection techniques to automate road damage identification from images, reducing manual inspection costs and improving detection accuracy.

Objective: Detect and classify road damage (potholes, cracks) using YOLOv8, output bounding boxes with class labels and confidence scores, count damage instances, and estimate severity levels.

2. Dataset Description

The dataset was sourced from Kaggle and contains annotated road damage images collected from multiple countries. It consists of 20 damage categories, with this project focusing on two primary classes: **crack** and **pothole**. Images were resized to 640×640 pixels for YOLOv8 compatibility.

Property	Details
Source	Kaggle – Road Damage Dataset (RDD)
Classes	2 classes: crack, pothole
Train Images	~2,660 images
Validation Set	666 images / 1,701 instances
Test Set	460 images
Image Size	640 × 640 px
Format	YOLO annotation format (.txt)

3. Method – YOLOv8

YOLOv8 (You Only Look Once, version 8) by Ultralytics is a single-stage object detector that divides the input image into a grid and simultaneously predicts bounding boxes, objectness scores, and class probabilities in a single forward pass. The YOLOv8s (small) variant was selected for its balance between speed and accuracy.

Key model characteristics:

- Architecture: 73 fused layers, 11.1M parameters, 28.4 GFLOPs
- Backbone: CSPNet with C2f modules for efficient feature extraction
- Optimizer: Adam | Learning Rate: 0.001 → 0.01
- Loss functions: Box loss (7.5), Class loss (0.5), DFL loss (1.5)
- Data Augmentation: random flip, HSV jitter, mosaic, erasing, scaling
- Hardware: NVIDIA Tesla T4 GPU (14 GB VRAM) via Google Colab

4. Experimental Work

Three experiments were conducted to evaluate the effect of the number of training epochs and data augmentation on model performance:

Experiment	Epochs	Augmentation	mAP50	mAP50-95	Precision	Recall
Exp 1 (Best)	50	Yes	85.4%	52.7%	81.7%	79.7%
Exp 2	30	Yes	83.9%	51.1%	83.4%	74.6%
Exp 3	30	No	83.9%	51.2%	83.2%	74.7%

* Green row = best performing experiment

Image Preprocessing

All images were resized to 640×640 pixels before training. The train/validation split follows the dataset's original partitioning (approximately 80% train / 20% validation).

5. Results

5.1 Per-Class Performance – Best Model (Experiment 1, 50 Epochs)

Class	Precision	Recall	mAP50	mAP50-95	Images	Instances
All (avg)	81.7%	79.7%	85.4%	52.7%	666	1,701
Crack	83.3%	86.7%	90.3%	57.9%	370	873
Pothole	80.1%	72.8%	80.5%	47.4%	296	828

Inference Speed: Preprocess: 2.2 ms | Inference: 10.4 ms | Postprocess: 2.2 ms per image (NVIDIA Tesla T4)

5.2 Confusion Matrices

Normalized confusion matrices show the proportion of correctly and incorrectly classified instances per class. Background false positives indicate cases where road damage was detected in non-damage regions.

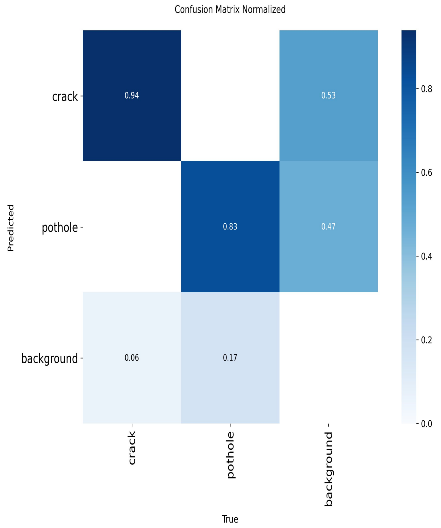

Figure 1: Experiment 1 (50 epochs, with augmentation)
Crack: 0.94 | Pothole: 0.83

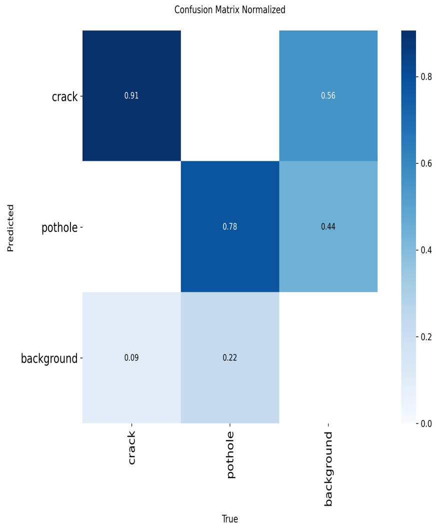

Figure 2: Experiment 2 (30 epochs, with augmentation)
Crack: 0.91 | Pothole: 0.78

5.3 Sample Detection Results

The following images show the model's output on test samples, with bounding boxes, class labels, and confidence scores for detected road damage.

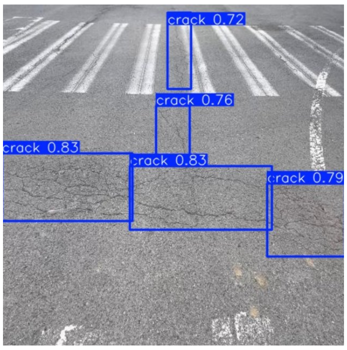

Fig. 3: Crack detection – multiple cracks detected (conf: 0.72–0.83)

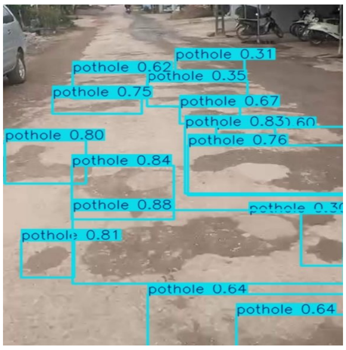

Fig. 4: Pothole detection – 13 potholes detected (conf: 0.31–0.88)

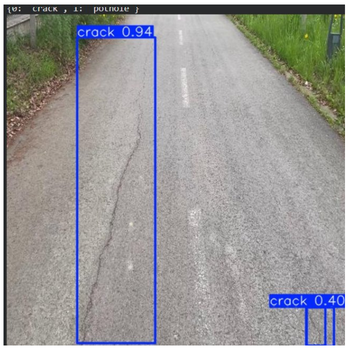

Fig. 5: Crack detection – high confidence crack (conf: 0.94)

Metric	Exp 1 (50 ep.)	Exp 2 (30 ep.)	Interpretation
Crack TP rate	0.94	0.91	Very high — model reliably detects cracks
Pothole TP rate	0.83	0.78	Good — minor false negatives from background
Crack→Background	0.06	0.09	Low miss rate for cracks
Pothole→Background	0.17	0.22	Some potholes missed (lower recall)

5.4 Training Progression

Training was monitored across all epochs. The model showed consistent improvement, starting from an mAP50 of 0.183 at epoch 1 and reaching 0.854 by epoch 50 (Experiment 1). Augmentation consistently provided a marginal improvement over training without augmentation (comparing Exp 2 vs Exp 3 at 30 epochs).

6. Discussion

- **Crack detection outperforms pothole detection** — The model achieves 94% true positive rate for cracks vs 83% for potholes. Cracks have more distinctive visual patterns, while potholes may appear similar to shadows or irregular road markings.
- **Augmentation improves generalization** — Comparing Exp 2 (with augmentation) and Exp 3 (without augmentation), both trained for 30 epochs, augmentation marginally boosts recall and mAP50-95 metrics.
- **More epochs improve performance** — Experiment 1 (50 epochs) achieves mAP50 = 85.4% vs Experiment 2 (30 epochs) at 83.9%, confirming that additional training yields better convergence.
- **Background confusion** — Some background regions are classified as crack (6%) or pothole (17%), suggesting the model occasionally misidentifies road texture or shadows as damage.
- **Fast inference speed** — At ~10.4 ms per image (Tesla T4), the model is suitable for real-time deployment on road inspection vehicles.

Best Configuration: YOLOv8s trained for **50 epochs with data augmentation** achieves the highest overall mAP50 of **85.4%**, Precision of **81.7%**, and Recall of **79.7%**. Crack detection (mAP50 = 90.3%) outperforms pothole detection (mAP50 = 80.5%) across all experiments.

7. Conclusion & Future Work

We introduced a YOLOv8-based road damage detection model capable of accurately identifying and classifying cracks and potholes in road images. The system outputs bounding boxes, class labels, and confidence scores, while also counting damage instances and estimating severity. The best model achieves an mAP50 of 85.4% after 50 training epochs with data augmentation.

Future Work:

- Increase the number of samples in the dataset to improve pothole recall.
- Try different object detection architectures such as Faster R-CNN, DETR, and RT-DETR.
- Expand classification to more damage types (longitudinal cracks, alligator cracks, rutting).
- Deploy the model on edge devices (Jetson Nano) for real-time road inspection.
- Integrate GPS tagging to map detected damage locations automatically.